Rational expressions

1 Fully simplify the following rational expressions:

a $\dfrac{49v}{35}$ **b** $\dfrac{15k}{20k}$ **c** $\dfrac{70v}{710}$ **d** $\dfrac{42xy}{18y}$

e $\dfrac{18p^7}{9p^3}$ **f** $\dfrac{30mn^2}{9m^2n}$ **g** $\dfrac{8k^2}{-12k}$ **h** $\dfrac{-12xy}{48x^2}$

2 For each of the following rational expressions:

i Factor the numerator and denominator.

ii Identify the greatest common factor between the numerator and denominator.

iii Fully simplify the rational expression.

a $\dfrac{4a}{4ab-20ac}$ **b** $\dfrac{2fg-6fh}{2f}$ **c** $\dfrac{6r+3rst}{3rs-12rt}$ **d** $\dfrac{4xy+8xy^2}{12x^3y^3}$

3 Fully simplify the following rational expressions:

a $\dfrac{6x-16}{12}$ **b** $\dfrac{x-4}{4-x}$ **c** $\dfrac{3y-45}{15-y}$ **d** $\dfrac{3x-12}{9x+15}$

e $\dfrac{3uv-12uw}{3u}$ **f** $\dfrac{4u}{4uv-20uw}$ **g** $\dfrac{2x+10y}{x+5y}$ **h** $\dfrac{3x-15y}{x-5y}$

4 Fully simplify the following rational expressions:

a $\dfrac{a^2-81}{9-a}$ **b** $\dfrac{x-5}{x^2-25}$ **c** $\dfrac{x-6}{x^2-36}$ **d** $\dfrac{m^2-1}{1+m}$

e $\dfrac{m^2-100}{m-10}$ **f** $\dfrac{50m^2+70mn}{80m^2}$ **g** $\dfrac{x-1}{x^2-2x+1}$ **h** $\dfrac{x-2}{x^2+4x-12}$

i $\dfrac{6(x+12)}{5x^2+120x+720}$ **j** $\dfrac{x^2+7x+10}{x+2}$ **k** $\dfrac{x^2+4x+4}{10(x+2)}$ **l** $\dfrac{2x^2+4x-198}{5(x-9)}$

5 Fully simplify the following rational expressions:

a $\dfrac{2r-8}{r^2-16}$ **b** $\dfrac{2x+8}{x^2+4x}$ **c** $\dfrac{20-5x}{2x^2-8x}$ **d** $\dfrac{15x-10}{3x^3-2x^2}$

e $\dfrac{12r+60}{3r^2-75}$ **f** $\dfrac{6(p-q)^2}{6p^2-6q^2}$ **g** $\dfrac{15-3xy}{x^2y-5x}$ **h** $\dfrac{12m^2+6mn}{18mn+36m^2}$

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6 Fully simplify the following rational expressions:

a $\dfrac{x^2+8x+16}{(x-5)(x+4)}$ b $\dfrac{(x-7)(x-1)}{x^2-9x+8}$ c $\dfrac{8x^2+2x-15}{(4x-5)^2}$ d $\dfrac{(3x+2)^2}{18x^2-3x-10}$

e $\dfrac{x^2-14x+49}{x^2-49}$ f $\dfrac{x^2+x-12}{x^2-3x}$ g $\dfrac{a^2+10a+24}{a^2-16}$ h $\dfrac{p^2+31p+220}{p^2-121}$

i $\dfrac{4x^2+12x+9}{9-4x^2}$ j $\dfrac{4x^2-3x-10}{4-x^2}$ k $\dfrac{x^2+2x-8}{8-2x^2}$ l $\dfrac{x^2-2x-8}{x^3-16x}$

7 Fully simplify the following rational expressions:

a $\dfrac{2x^2+10x-100}{2x+20}$ b $\dfrac{9x^2+9x}{10x^2+20x+10}$ c $\dfrac{x^2+14x+49}{x^2+4x-21}$ d $\dfrac{x^2-10x+25}{x^2+5x-50}$

e $\dfrac{4x^2+8x-32}{x^2+3x-10}$ f $\dfrac{x^3-x^2-20x}{x^2+7x+12}$ g $\dfrac{25-16x^2}{4x^2+7x-15}$ h $\dfrac{x^2+x-12}{5x^2+18x-8}$

i $\dfrac{8x^2-18x+10}{4x^2+x-5}$ j $\dfrac{2x^2-7x+3}{4x^2-17x+15}$ k $\dfrac{2x^2-13x+15}{2x^2-x-3}$ l $\dfrac{4x^2+7x-2}{6+x-x^2}$

8 Luciana has attempted to fully simplify a rational expression and showed her work:

1 $\dfrac{2x+7}{2x^2+6x} = \dfrac{2x+7}{2x(x+3)}$ Factor the denominator

2 $= \dfrac{7}{x+3}$ Divide out $2x$ from the numerator and denominator

Identify where Luciana has made an error and explain what it is.

9 Qin has attempted to fully simplify a rational expression and showed his work:

1 $\dfrac{8x^2-24x}{x^2+5x-24} = \dfrac{8x(x-3)}{(x+8)(x-3)}$ Factor the numerator and denominator

2 $= \dfrac{8x}{x+8}$ Divide out $(x+3)$ from the numerator and denominator

3 $= \dfrac{8x}{x} + \dfrac{8x}{8}$ Additive property of fractions

4 $= 8 + x$ Simplify the fractions

Identify where Qin has made an error and explain what it is.

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10 For each of the following, find the value(s) of k that would allow the rational expression to be simplified into a linear expression:

a $\dfrac{kx^2 + 12x}{4x}$ **b** $\dfrac{x^2 + kx + 42}{x + 6}$ **c** $\dfrac{9x^2 + 6x + k}{3x}$ **d** $\dfrac{x^2 - 7x - 18}{x + k}$

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